MATH 310
Identifying Musical Instruments with Mathematics

Core Equations and Theoretical Fou)ndation

1. **Euler's Formula**

$$e^{-i\omega n} = \cos \omega n - i \sin \omega n$$

Links circular motion in the complex plane to sinusoidal oscillations; fundamental to all Fourier analysis.

2. **Discrete Fourier Transform (DFT)**

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{-j\frac{2\pi}{N}nk}, \quad k = 0, 1, \ldots, N-1$$

Decomposes a finite discrete signal into complex exponential components. Invertible: $x[n] = \dfrac{1}{N}\displaystyle\sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi}{N}nk}$.

3. **Poisson Summation Formula (Sampling Theory)**

$$\sum_{k=-\infty}^{\infty} X_c\left(f - kf_s\right) = \frac{1}{T_s}\sum_{n=-\infty}^{\infty} x_c(nT_s)e^{-j2\pi f n T_s}$$

[Relates continuous signal spectrum to discrete sampled signal; explains aliasing and Nyquist bound]

4. **Short-Time Fourier Transform (STFT)**

$$X_{\text{STFT}}(t, f) = \int_{-\infty}^{\infty} x(\tau) \cdot w(\tau - t) \cdot e^{-j2\pi f\tau}\, d\tau$$

Discrete form: $X_{\text{STFT}}(m, k) = \displaystyle\sum_{n=0}^{N-1} x[mH + n] \cdot w[n] \cdot e^{-j\frac{2\pi}{N}nk}$, where m is frame index and H is hop length.
[Captures time-varying frequency content via sliding window]

Part I: Digital Signal Processing Foundations

1. Sampling Theory and the Nyquist-Shannon Theorem

Definition 1 (**Bandlimited Signal**). *A continuous signal $x_c(t)$ is bandlimited if its Fourier transform $X_c(f) = 0$ for all $|f| > B$, where B is the bandwidth.*

Theorem 2 (**Nyquist-Shannon Sampling Theorem**). *Let $x_c(t)$ be a bandlimited signal with bandwidth B. If $x_c(t)$ is sampled at rate $f_s \geq 2B$ (equivalently, sampling period $T_s \leq \dfrac{1}{2B}$), then $x_c(t)$ can be perfectly reconstructed from its samples via:*

$$x_c(t) = \sum_{n=-\infty}^{\infty} x_c(nT_s) \cdot sinc\left(\frac{t - nT_s}{T_s}\right)$$

where $sinc(u) = \dfrac{\sin(\pi u)}{\pi u}$.

Proof Sketch: Under the Poisson Summation Formula, periodic repetition of $X_c(f)$ at multiples of f_s does not overlap if $f_s > 2B$ (no aliasing). An ideal lowpass filter at cutoff $f_s/2$ (the Nyquist frequency) recovers $X_c(f)$ exactly, and the inverse Fourier transform yields the reconstruction formula.

Application to Audio:

1. CD-quality audio uses $f_s = 44,100$ Hz $\Rightarrow$ Nyquist frequency $f_N = 22,050$ Hz
2. Human hearing range: approximately 20 Hz to 20,000 Hz
3. A pre-sampling anti-aliasing filter (analog lowpass, cutoff ≈ 20 kHz) ensures no information loss
4. Any frequency above f_N will fold back (alias) into the range $[0, f_N]$, corrupting the signal

2. Quantization and Bit Depth

Definition 3 (**Quantization Error**). *When a continuous amplitude value $x(t)$ is mapped to one of $L = 2^B$ discrete levels (where B is bit depth), the quantization error is $e[n] = x_q[n] - x[n]$, where x_q is the quantized value.*

For uniform quantization with step size $\Delta = \dfrac{V_{\max}}{2^B}$ (full-scale voltage divided by number of levels):

1. Quantization error is bounded: $|e[n]| \leq \dfrac{\Delta}{2}$
2. Assuming white noise model, quantization noise power: $\sigma_q^2 = \dfrac{\Delta^2}{12}$
3. Signal-to-Quantization-Noise Ratio (SQNR):
$$\text{SQNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} \approx 6.02B + 1.76 \text{ dB}$$
Each additional bit improves SQNR by approximately 6 dB.
4. 16-bit audio: $2^{16} = 65,536$ levels, SQNR ≈ 98 dB
5. 24-bit audio: $2^{24} = 16,777,216$ levels, SQNR ≈ 146 dB

Why Higher Bit Depth Matters: Higher-fidelity instruments have subtle harmonic components with dynamic range exceeding 96 dB. 24-bit audio preserves these nuances, improving feature extraction and classification.

3. Frequency Resolution and DFT Analysis

Definition 4 (**Frequency Resolution**). *Given a DFT of N samples at sampling rate f_s, the frequency resolution (bin spacing) is:*
$$\Delta f = \frac{f_s}{N} \textit{ Hz/bin}$$

Interpretation:

1. A DFT output provides spectral estimates at N discrete frequencies: $f_k = k \cdot \Delta f$ for $k = 0, 1, \ldots, N-1$
2. Finer resolution (smaller Δf) requires larger N (longer time window)
3. Trade-off: Larger N improves frequency resolution but worsens time resolution (we average over longer duration)
4. Example: $f_s = 44,100$ Hz, $N = 2,048$ samples $\Rightarrow \Delta f = 21.53$ Hz, time window $= 46.4$ ms
5. Example: $f_s = 44,100$ Hz, $N = 4,096$ samples $\Rightarrow \Delta f = 10.77$ Hz, time window $= 92.9$ ms

Leakage and the Picket-Fence Effect: If a true frequency f_0 does not align with a DFT bin center, spectral energy spreads across adjacent bins (leakage), distorting magnitude estimates.

Part II: Windowing and Spectral Leakage

1. The Spectral Leakage Problem

Definition 5 (Spectral Leakage). *When a finite frame of audio does not contain an integer number of periods, the DFT implicitly assumes periodic repetition, creating artificial discontinuities at the frame boundary. These discontinuities introduce unwanted high-frequency components across the spectrum.*

Mathematical Root Cause: The DFT of a windowed signal $y[n] = x[n] \cdot \mathbb{1}_{[0,N-1]}[n]$ is the convolution of $X[k]$ (true spectrum) with $W[k]$ (spectrum of the rectangular window):

$$Y[k] = \frac{1}{N} \sum_{m=0}^{N-1} X[m] \cdot W[k-m]$$

The rectangular window has high side lobes (≈ -13 dB), causing significant leakage.

2. Window Functions: Design and Trade-offs

Definition 6 (Window Function). *A window $w[n]$ is a smooth function on $[0, N-1]$ with $w[0] = w[N-1] = 0$ (or close to zero), tapered at the edges. Multiplying the signal by $w[n]$ before DFT reduces leakage.*

Key Parameters:

1. **Main Lobe Width:** Affects frequency resolution (narrower is better)
2. **Side Lobe Attenuation:** Attenuates leakage into distant bins (higher is better)
3. **Asymptotic Rolloff:** How quickly side lobes decay with bin distance

Common Windows:

1. **Rectangular Window:** $w[n] = 1$ for all $n \in [0, N-1]$
   - Main lobe width: $\Delta f = \dfrac{4\pi}{N}$
   - Side lobe level: -13 dB (poorest)
   - Asymptotic rolloff: -6 dB/octave
2. **Hann Window:**
$$w[n] = 0.5\left(1 - \cos\left(\frac{2\pi n}{N-1}\right)\right), \quad n = 0, 1, \ldots, N-1$$
   - Main lobe width: $\Delta f = \dfrac{8\pi}{N}$
   - Side lobe level: -32 dB (excellent leakage suppression)
   - Asymptotic rolloff: -18 dB/octave
   - Best for general-purpose spectral analysis
3. **Hamming Window:**
$$w[n] = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$
   - Main lobe width: $\Delta f = \dfrac{8\pi}{N}$ (same as Hann)
   - Side lobe level: -43 dB at first side lobe (better than Hann asymptotically)

- Asymptotic rolloff: -6 dB/octave (weaker rolloff far out)
- Practical choice: balances main lobe and near side lobes

4. **Blackman Window:**

$$w[n] = 0.42 - 0.5\cos\left(\frac{2\pi n}{N-1}\right) + 0.08\cos\left(\frac{4\pi n}{N-1}\right)$$

- Main lobe width: $\Delta f = \dfrac{12\pi}{N}$ (widest, loses frequency resolution)
- Side lobe level: -58 dB (superior leakage suppression)
- Asymptotic rolloff: -18 dB/octave
- Use when spectral leakage is critical; accept frequency resolution loss

Recommendation: For instrument timbre analysis, the Hann window offers excellent leakage suppression with minimal frequency resolution loss. Use a frame length of 2,048 to 4,096 samples.

3. Short-Time Fourier Transform (STFT) and Spectrograms

Definition 7 (STFT). *The STFT is defined as:*

$$X_{STFT}(m,k) = \sum_{n=0}^{N-1} x[mH+n] \cdot w[n] \cdot e^{-j\frac{2\pi}{N}nk}$$

where m indexes frames, H is the hop length (frame advance), $w[n]$ is the window function, and k indexes frequency bins.

Key Parameters:

1. **Frame Length N:** Typically 2,048 or 4,096 samples
2. **Hop Length H:** Typically $N/4$ (75% overlap) for smooth time-frequency representation
3. **Window Overlap:** Higher overlap (e.g., 75%) preserves information better than lower overlap (e.g., 50

Interpretation: The spectrogram $|X_{\text{STFT}}(m,k)|^2$ is a 2D time-frequency power distribution. Rows represent frequency content; columns represent time evolution. For a musical note:

1. **Attack:** Rapid spectral onset (transient)
2. **Decay:** Evolution of harmonics (sustain phase)
3. **Release:** Fade-out of energy

Different instruments exhibit distinct temporal signatures, enabling visual classification and feature design.

Part III: Timbre and Harmonic Analysis

1. The Harmonic Series and Instrument Timbre

Definition 8 (Harmonic Series). *For a vibrating object with fundamental frequency f_0, the harmonic series is:*

$$f_k = k \cdot f_0, \quad k = 1, 2, 3, \ldots$$

Each harmonic is a standing wave at an integer multiple of the fundamental.

Definition 9 (Timbre). *Timbre is the perceived quality of a sound, determined by the relative amplitudes and phases of the harmonic components. Mathematically, timbre signature is encoded in the spectral envelope: the smooth curve connecting the peaks of harmonic magnitudes.*

Harmonic Content Patterns:

1. **Trumpet:** Emphasizes many high harmonics (up to 15th+) $\Rightarrow$ bright, brassy timbre
2. **Flute:** Emphasizes fundamental and first few harmonics (weak 5th+) $\Rightarrow$ mellow, flute-like timbre
3. **Clarinet:** Emphasizes odd harmonics (1st, 3rd, 5th, ...) due to cylindrical bore $\Rightarrow$ warm, woody timbre
4. **Cello:** Rich harmonics in mid range, weak very-high harmonics $\Rightarrow$ deep, sonorous timbre
5. **Piano:** Non-harmonic overtones (inharmonicity) due to string stiffness $\Rightarrow$ unique, complex spectrum

Inharmonicity: For stiff strings (pianos, harps), overtone frequencies deviate from exact harmonicity:

$$f_k \approx k f_0 \sqrt{1 + \epsilon k^2}$$

where ϵ is the inharmonicity coefficient. This creates a characteristic "bell-like" quality that distinguishes pianos from simpler instruments.

2. Feature Extraction: Spectral Signatures

All features below are computed per STFT frame, then aggregated (mean, std, median) over the entire audio file to form a feature vector.

1. **Spectral Centroid:**

$$\mathrm{SC} = \frac{\sum_{k=0}^{N-1} f_k \cdot |X[k]|}{\sum_{k=0}^{N-1} |X[k]|}$$

   Measures the "center of mass" of the power spectrum. Interpretations:

   - High SC (> 3 kHz): Bright instruments (trumpet, violin, cymbals)
   - Low SC (< 1 kHz): Dark instruments (cello, French horn, tuba)
   - Mid SC (1–3 kHz): Warm instruments (flute, clarinet, vocals)

2. **Spectral Rolloff:**

$$f_{\text{rolloff}} = \arg\min_f \left(\sum_{k: f_k < f} |X[k]|^2 > 0.85 \sum_{k=0}^{N-1} |X[k]|^2 \right)$$

   Frequency below which 85% of spectral energy is concentrated. High rolloff indicates presence of high-frequency harmonics; low rolloff indicates noisy or dull instruments.

3. **Spectral Flux:**

$$\mathrm{SF}(m) = \sum_{k=0}^{N-1} (|X_{\text{STFT}}(m,k)| - |X_{\text{STFT}}(m-1,k)|)^2$$

   Measures frame-to-frame change in spectrum. Detects onsets and transients. High flux $\Rightarrow$ rapid spectral change (piano attack, drum strike); low flux $\Rightarrow$ stable spectrum (sustained violin note).

4. **Zero-Crossing Rate (ZCR):**

$$\mathrm{ZCR}(m) = \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{1}\{x[mH+n] \cdot x[mH+n+1] < 0\}$$

Counts sign changes in time-domain signal. Interpretations:

- High ZCR (> 0.3): Noisy/unvoiced components (cymbals, snare, white noise)
- Low ZCR (< 0.1): Harmonic/tonal (violin, flute, voice)

5. **Mel-Frequency Cepstral Coefficients (MFCCs):**

   (a) Convert frequency scale to Mel scale (logarithmic, matching human perception):

$$f_{\mathrm{mel}} = 2595 \log_{10}\left(1 + \frac{f}{700}\right)$$

   (b) Create mel-spaced filterbank: 40 triangular filters covering 0 to 8 kHz

   (c) Compute log-magnitude of mel-filtered spectrum: $\log(E_k)$ for each filter k

   (d) Apply Discrete Cosine Transform (DCT):

$$c_i = \sum_{k=1}^{M} \log(E_k) \cos\left(\frac{\pi i(k-0.5)}{M}\right), \quad i = 0, 1, \ldots, 12$$

   (e) Extract first 13 coefficients (MFCCs 0–12); omit MFCC-0 (energy) or include as separate feature

   MFCCs capture perceptually meaningful spectral shape; highly effective for timbre classification.

6. **Spectral Contrast:**

$$\mathrm{SC}_b = \mathrm{mean}(|X[k]|) - \mathrm{mean}(|X[k]| \text{ for valleys in } b\text{-th band})$$

   Measures peaks minus valleys in each frequency band. Captures harmonic structure; useful for discriminating pitched vs. noisy instruments.

Part IV: Classification Methods

1. Support Vector Machine (SVM)

Definition 10 (SVM: Linear Case). *Given training data $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$ where $\mathbf{x}_i \in \mathbb{R}^d$ and $y_i \in \{-1, +1\}$, the SVM finds a separating hyperplane by solving:*

$$\min_{\mathbf{w}, b, \boldsymbol{\xi}} \frac{1}{2}\|\mathbf{w}\|^2 + C \sum_{i=1}^{m} \xi_i$$

subject to $y_i(\mathbf{w}^T \phi(\mathbf{x}_i) + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$.

Key Concepts:

1. **Margin:** Distance from hyperplane to nearest training point. Larger margin $\Rightarrow$ better generalization
2. **Kernel Trick:** Non-linear classification via implicit mapping ϕ without computing it explicitly
3. **RBF Kernel:** $K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ handles non-linear decision boundaries
4. **Regularization Parameter C:** Trade-off between margin maximization and training error minimization

For Multi-class Instrument Classification: Use One-vs-Rest (OvR) encoding: train K binary classifiers (one per instrument class) and select the class with highest decision function value.

2. Random Forest Classifier

Ensemble Decision Trees:

1. Train T decision trees on random subsets (with replacement) of training data
2. At each split, consider random subset of features
3. Prediction for new sample: majority vote across all T trees

Advantages:

- Handles high-dimensional mixed feature types naturally
- Provides feature importance via average decrease in impurity
- Robust to overfitting (via averaging); typically performs well with little tuning

3. k-Nearest Neighbors (k-NN)

Algorithm: For a query point $\mathbf{x}_q$, find the k nearest training samples (via Euclidean distance) and assign the majority class among them.

Distance Metric: Use standardized Euclidean distance to account for varying feature scales:

$$d(\mathbf{x}_i, \mathbf{x}_j) = \sqrt{\sum_{d=1}^{D} \frac{(x_i^{(d)} - x_j^{(d)})^2}{\sigma_d^2}}$$

Hyperparameter Selection: Typical values $k \in \{3, 5, 7\}$. Use cross-validation to select optimal k.

Part V: Evaluation and Validation

1. Confusion Matrix and Metrics

Definition 11 (**Confusion Matrix**). *For K instrument classes, a $K \times K$ matrix $\mathbf{C}$ where C_{ij} = number of samples from class i predicted as class j. Diagonal entries represent correct predictions.*

Per-class metrics (for class i):

1. **True Positive (TP):** C_{ii} (correctly identified as class i)
2. **False Positive (FP):** $\sum_{j \neq i} C_{ji}$ (misclassified as class i)
3. **False Negative (FN):** $\sum_{j \neq i} C_{ij}$ (class i samples misclassified as others)
4. **Precision:** $P_i = \dfrac{\text{TP}}{\text{TP} + \text{FP}}$ — "Of samples predicted as class i, what fraction is correct?"
5. **Recall:** $R_i = \dfrac{\text{TP}}{\text{TP} + \text{FN}}$ — "Of actual class i samples, what fraction did we find?"
6. **F1-Score:** $F_i = 2\dfrac{P_i \cdot R_i}{P_i + R_i}$ — harmonic mean of precision and recall

Macro-Averaging (unweighted):

$$\text{Accuracy} = \frac{\sum_{i=1}^{K} C_{ii}}{\sum_{i,j} C_{ij}}, \quad \text{Macro-F1} = \frac{1}{K} \sum_{i=1}^{K} F_i$$

2. Cross-Validation and Overfitting Detection

Definition 12 (**k-Fold Cross-Validation**). *Partition data into k disjoint folds. For each fold i:*

1. *Train on folds $1, \ldots, i-1, i+1, \ldots, k$ (combined)*
2. *Validate on fold i*
3. *Record validation metric*

Report mean and standard deviation of metrics across all folds.

Typical k values: 5-fold or 10-fold cross-validation. Stratified cross-validation (preserve class distribution in each fold) is recommended for imbalanced datasets.

Overfitting Detection: If training accuracy $\gg$ validation accuracy, the model is overfitting. Remedies:

- Reduce model complexity (fewer features, smaller tree depth)
- Increase regularization (increase SVM parameter C, or increase Random Forest min_samples_leaf)
- Collect more training data

Part VI: Project Workflow

1. **Data Preparation:** Collect audio (NSynth, IRMAS) with known labels; split 70%/30% train/test or use cross-validation
2. **Preprocessing:**
   - Verify sampling rate (f_s typically 44,100 or 48,000 Hz)
   - Normalize amplitude to $[-1, 1]$ range
   - Remove silence (zero-padding and silence detection)
3. **Feature Extraction:**
   - Compute STFT with Hann window, 2,048 samples, 50% overlap
   - For each frame: compute spectral centroid, rolloff, flux, ZCR, MFCCs (13 coefficients)
   - Aggregate: mean, std, median of each feature across all frames
   - Result: $\approx$ 40–50 features per audio file
4. **Normalization:** Standardize each feature to zero mean and unit variance (important for distance-based classifiers)
5. **Model Training:**
   - Train SVM (RBF kernel, $\gamma = 1/d$, tune C via grid search)
   - Train Random Forest (500 trees, max_depth = 15, min_samples_leaf = 5)
   - Train k-NN ($k = 5$)
6. **Validation:** 5-fold cross-validation on training set to estimate generalization error
7. **Testing:** Evaluate on held-out test set; compute accuracy, F1-score, per-class precision/recall
8. **Analysis:**
   - Plot confusion matrix

- Visualize spectrograms for each instrument class
- Extract Random Forest feature importance (which features discriminate best?)
- Analyze misclassified samples (why are violin and flute confused?)

9. **Documentation:** Report methodology, results, limitations, and future improvements

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